

Sample End Semester Question Paper - 2025

Subject: Control Systems | Code: EE302 | Semester: 3

Branch: EEE | Regulation: R20 (sample) | Time: 3 Hours | Max Marks:

This is a SAMPLE question paper generated for Backlog Buddy (demo/testing data).
It is not an official university paper and is provided to demonstrate the platform.

PART A - Short questions (10 x 1 = 10 Marks)

1. Define gain margin.
2. What is a PID controller?
3. What is steady-state error?
4. Define damping ratio.
5. What is a block diagram?
6. Define a control system.
7. What is a transfer function?
8. Name two time domain specifications.
9. What is the Routh array?
10. What is a root locus?

PART B - Long questions (7 x 11 = 77 Marks)

1. Explain steady-state error analysis and the effect of system type.
2. Explain PID controllers and their effect on system performance.
3. Find the transfer function of a given mechanical or electrical system.
4. Explain time response of second-order systems: rise time, peak time, overshoot and settling time.
5. Explain the Routh-Hurwitz criterion. Test stability of a given system.
6. Draw and explain the root locus of a given system.
7. Explain Bode plots. Find gain and phase margins.